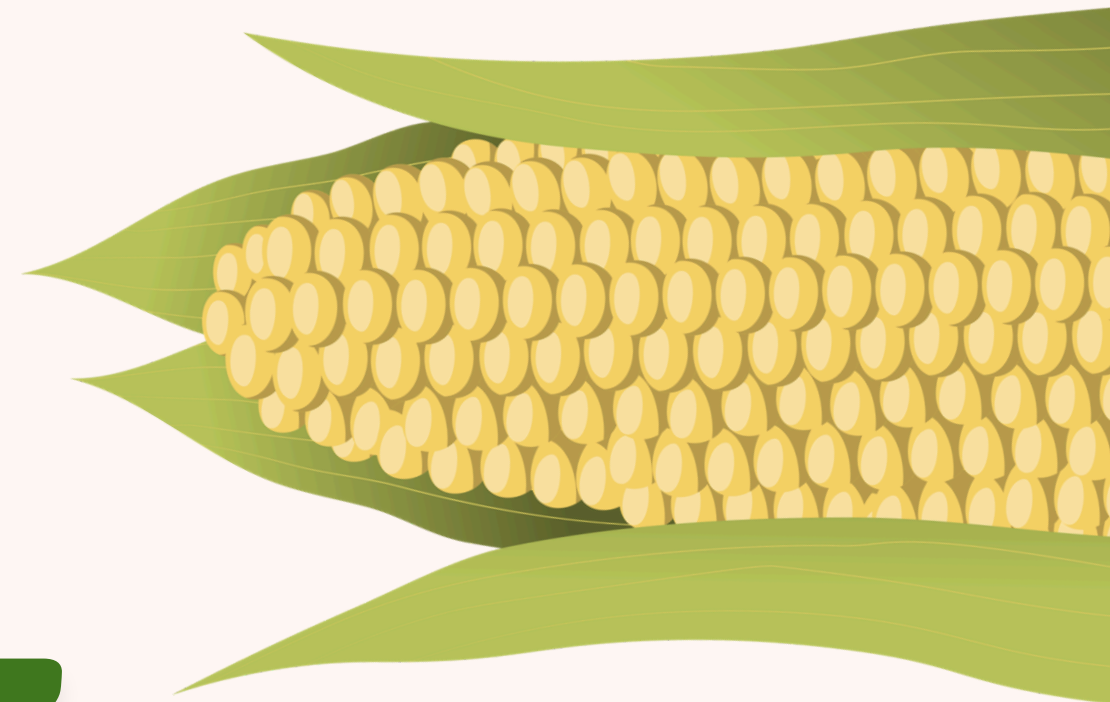




FUNGICIDE EXPLORATION

The Effects of Different Fungicides on the Yields of
Corn Crops
BIS15L | Group 11





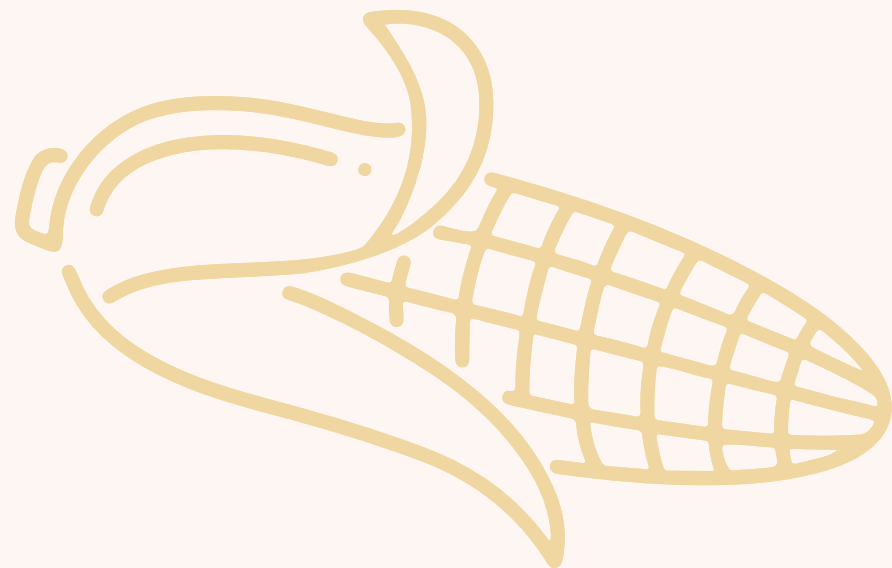
Our Team

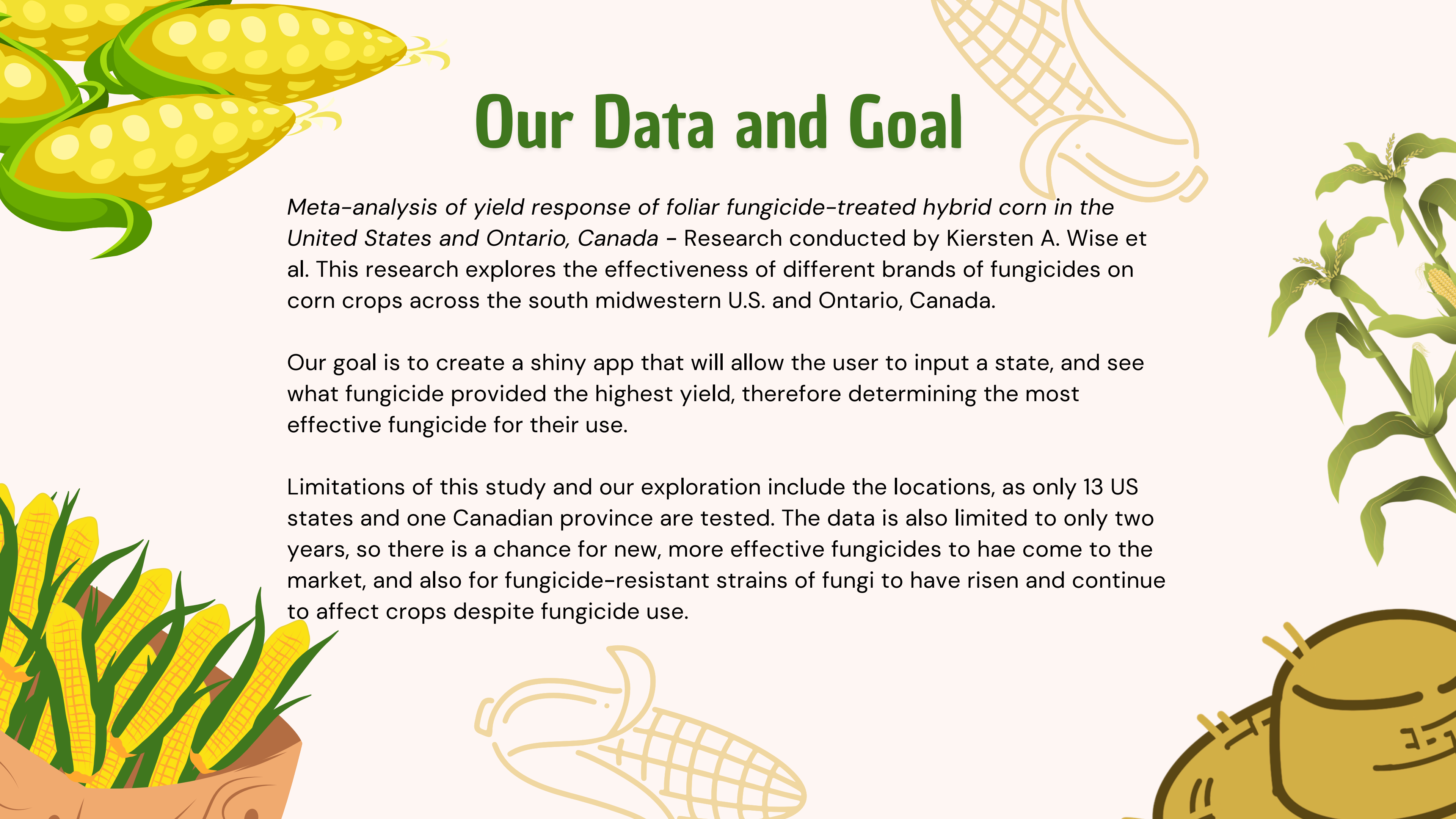


Vaishnavi Maganti

Alejandro Acevedo

Catalina De Amorrortu



The slide is decorated with various corn-related illustrations. In the top left, there are three realistic-looking yellow corn cobs with green husks. In the top right, there is a faint, stylized outline of a corn cob. On the right side, there is a green corn plant with several leaves and a small ear of corn. In the bottom left, there is a basket filled with several yellow corn cobs. In the bottom center, there is a faint, stylized outline of a corn cob. In the bottom right, there is a large, golden-yellow cornucopia or sack with a drawstring top, overflowing with corn.

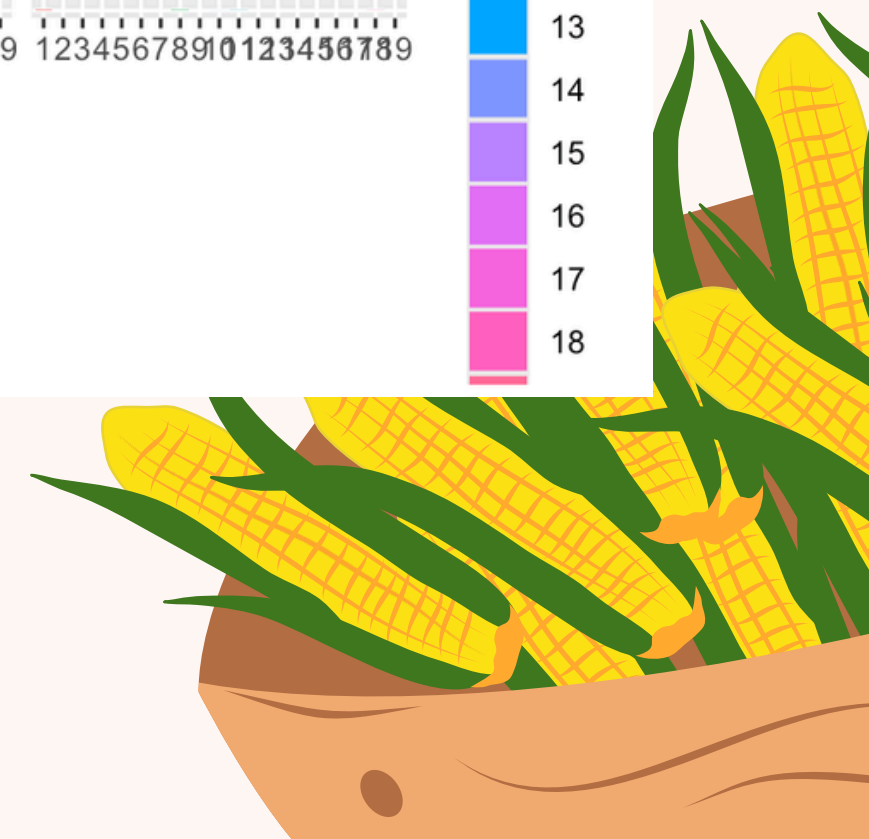
Our Data and Goal

Meta-analysis of yield response of foliar fungicide-treated hybrid corn in the United States and Ontario, Canada – Research conducted by Kiersten A. Wise et al. This research explores the effectiveness of different brands of fungicides on corn crops across the south midwestern U.S. and Ontario, Canada.

Our goal is to create a shiny app that will allow the user to input a state, and see what fungicide provided the highest yield, therefore determining the most effective fungicide for their use.

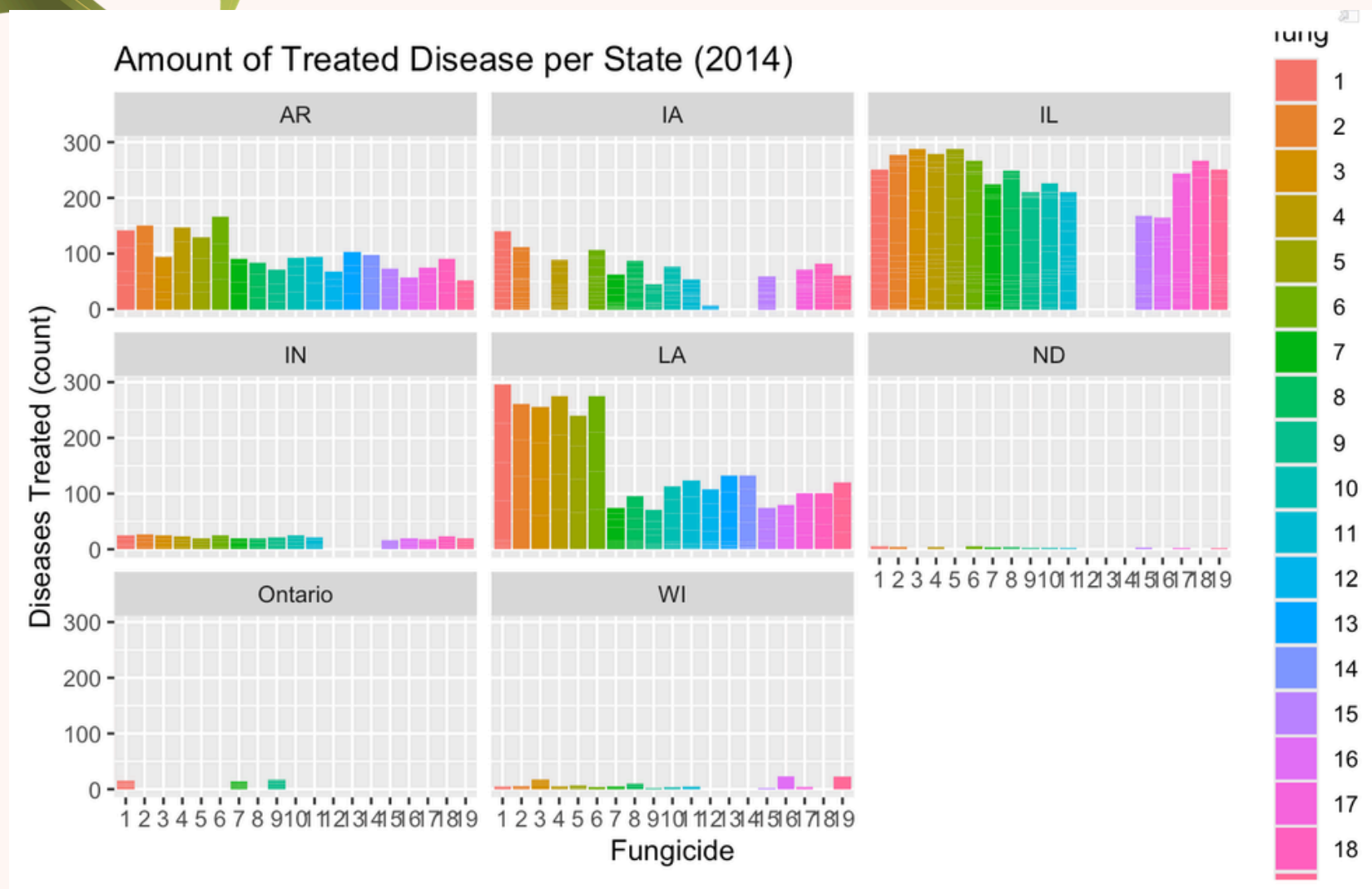
Limitations of this study and our exploration include the locations, as only 13 US states and one Canadian province are tested. The data is also limited to only two years, so there is a chance for new, more effective fungicides to have come to the market, and also for fungicide-resistant strains of fungi to have risen and continue to affect crops despite fungicide use.

- Fungicide effect

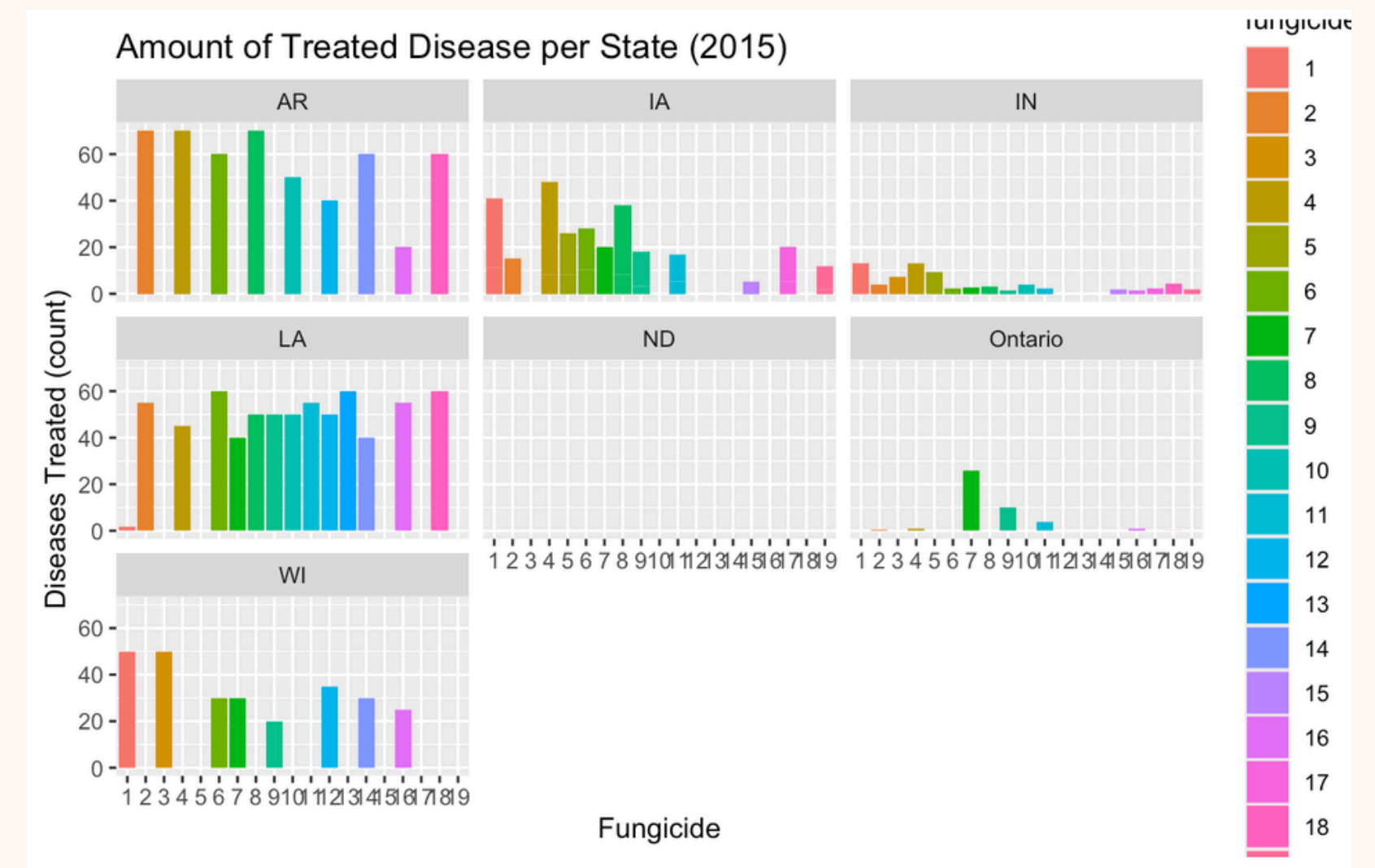


Data

- Fungicide resistance
 - Comparing the change in the total disease by fungicide and state in 2014 and 2015, we can determine if any fungicide-resistant strains appeared



2014

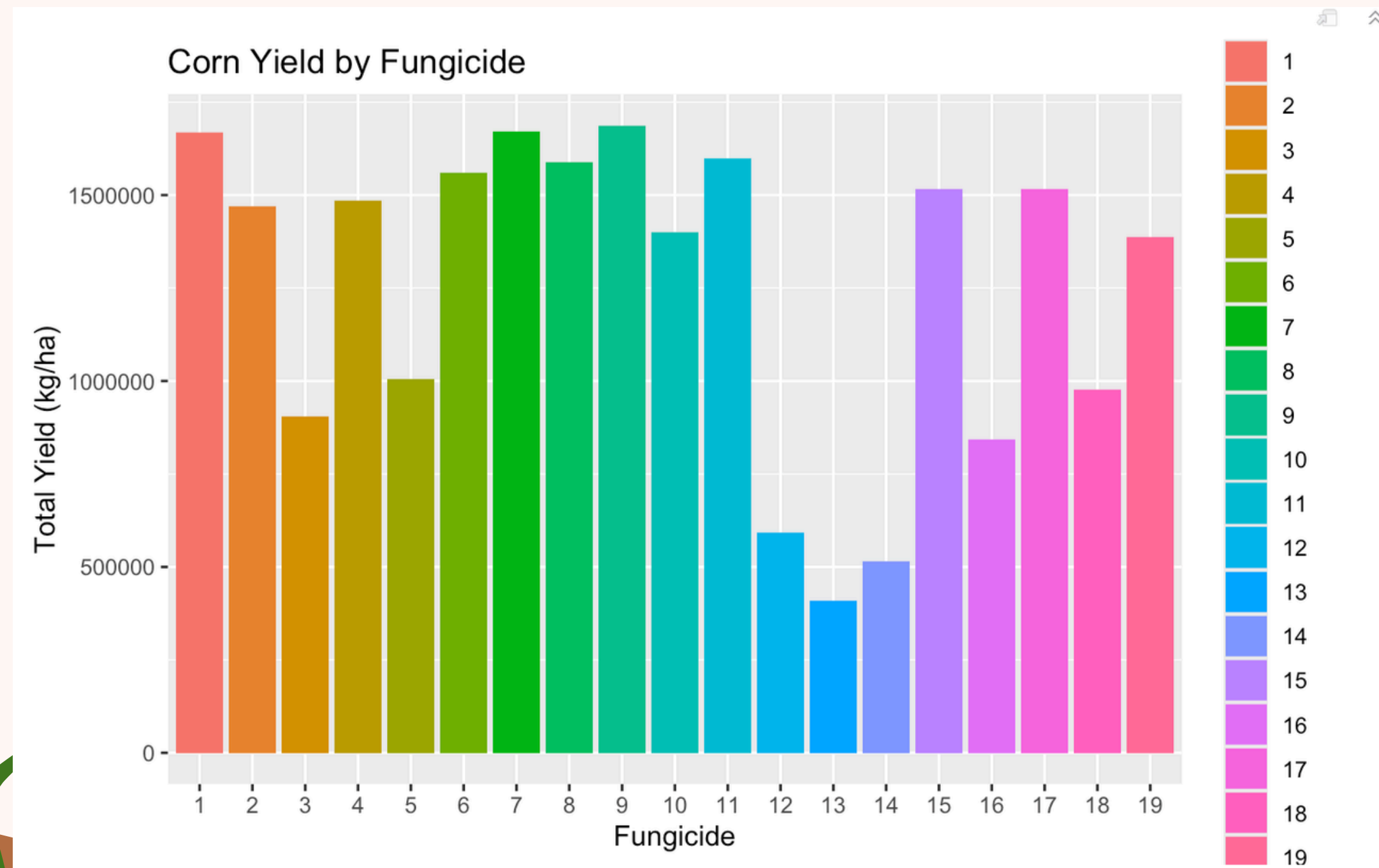


2015

IL was not tested in 2015, so it has been removed from the 2015 graphs.

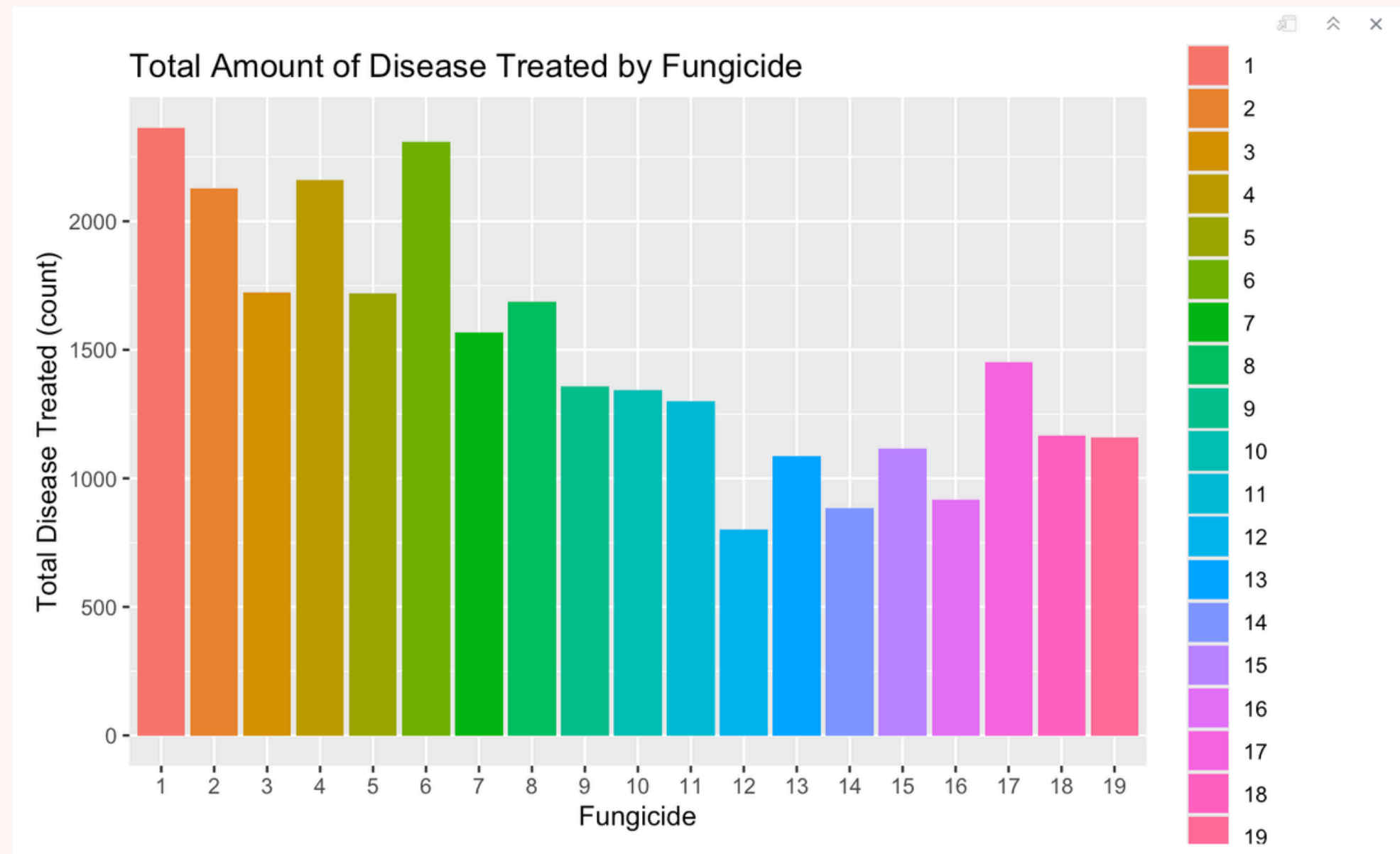
Data

- Overall, which fungicide provides us with the highest yield of corn?



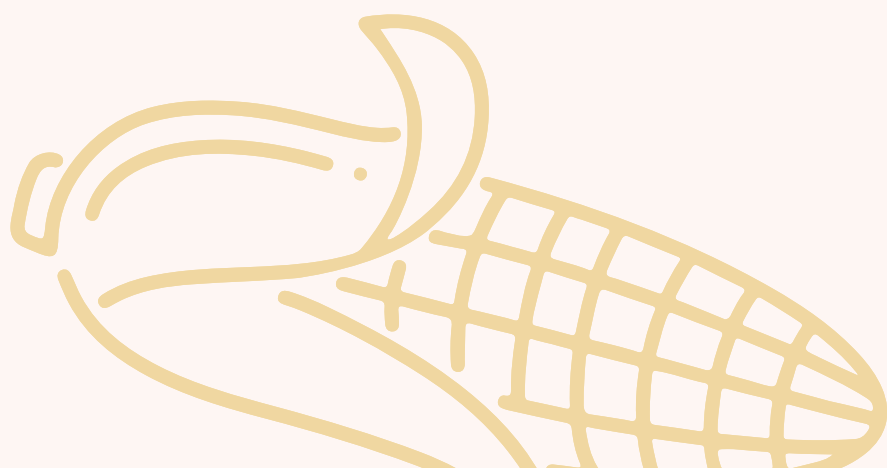
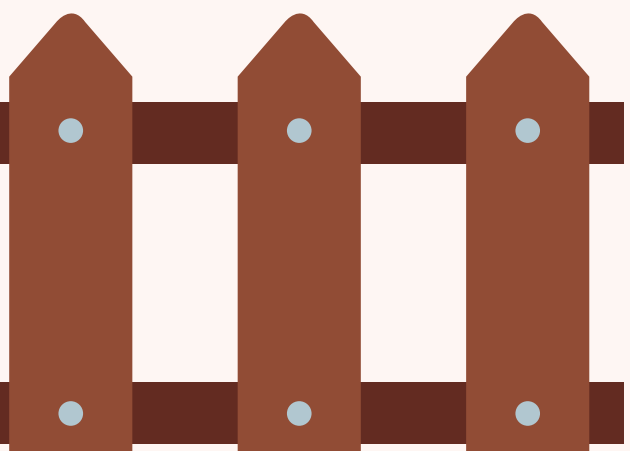
Data

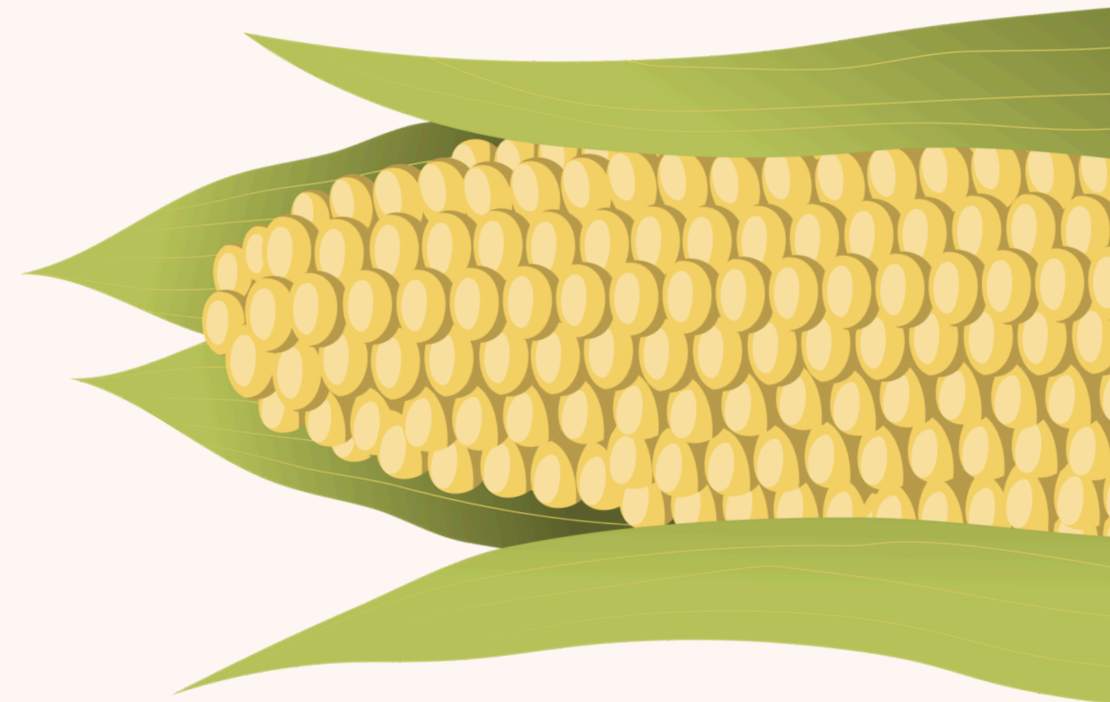
- Overall, which fungicide treats the highest amount of diseases/fungi?





Shiny App





Thank You

